

us System Setup Procedure (English Translation)

1) Install SQL Server 2022 Express

Run SETUP.EXE → Run as Administrator → Select “**New SQL Server stand-alone installation**” → Accept the license terms → Next → Do not check “Use Microsoft Update to check for updates” → Next → Next
Uncheck “**Azure extension for SQL Server**” → Next Leave the instance features as default
→ Next Select Default Instance → Next Service accounts → leave as default → Next
Select **Mixed Mode Authentication** Set password: **test@123** → Next Make sure to write down this password.

2) Install SQL Server Management Studio (SSMS)

Download: <https://learn.microsoft.com/ja-jp/ssms/install/install>

Run vs_SSMS.exe → Run as Administrator → Continue → On the workload screen, do **not** select anything → Install

When SSMS starts, skip the account setup (you can add it later).

Connection settings:

Server name: (local)

Authentication: SQL Server Authentication

User name: sa

Password: test@123

Check **“Remember password”**

Database: <Default>

Encryption: Optional

If Object Explorer appears, the installation is successful.

3) Restore the Database

In Object Explorer: → Right-click **Databases** → Select **Restore Database** → Choose **Device** → click ...

Select:

コード

C:\Weather\DBBackup\weather_202612.bak

→ OK → OK

If the message **“Database 'weather' restored successfully.”** appears, the restore is complete.

4) Edit C:\weather¥config¥dbconfig

Open the file with Notepad. Modify the following fields:

Server

Password

OpenWeatherApiKey

Example:

json

```
{  
  "Server": "IP-ADDRESS",  
  "Database": "weather",  
  "User": "sa",  
  "Password": "test@123",  
  "OpenWeatherApiKey": "?????????"  
}
```

If IP-ADDRESS does not work, try using (local).

5) Verify Data Retrieval (Manual Test)

Command Prompt:

コード

```
cd C:\weather\script
```

```
WeatherLogger.bat
```

If “**1**” continues to appear, data is being inserted into the database.

SQL check:

```
sql
```

```
SELECT *
```

```
FROM [weather].[dbo].[WeatherLog]
```

```
ORDER BY 1 DESC;
```

6) Add Windows Task (Automatic Data Retrieval)

Create a new task: **Name:** get-temp-data

General

Run whether user is logged on or not

Run with highest privileges

Triggers

Daily at: **0:00, 4:00, 8:00, 12:00, 14:00, 16:00, 20:00**

Actions

Program:

コード

C:\weather\Script\WeatherLogger.bat

Start in:

コード

C:\weather\Script

Settings

Uncheck **“Stop the task if it runs longer than…”**

After registering the task, click **“Enable All Tasks History”** on the right side. “History (disabled)” will change to “History”.

7) Install Python (for viewing stored data)

Run:

コード

```
python-3.14.6-amd64.exe
```

→ Run as Administrator Check:

Use admin privileges when installing py.exe

Add python.exe to PATH

Click **Install Now**.

8) Install Python ODBC Driver

PowerShell:

コード

```
pip install pyodbc
```

If you see:

コード

Successfully installed pyodbc

the installation is complete.

9) View Retrieved Data

Command Prompt:

コード

```
C:\weather\Script>weather_viewer.py
```

10) Automate DB Backup

Create another Windows Task following step **6**, but set the program to your DB backup script.